Dynamics of cell death due to N and P starvation across intra-clade diversity in *Prochlorococcus*

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Abstract

Nutrient starvation and subsequent mortality are processes that can shape ecosystem dynamics and influence global biogeochemical cycles yet are poorly understood. Here, we examined the dynamics of culture decline in 15 strains of *Prochlorococcus*, globally abundant marine cyanobacteria, under nitrogen (N) and phosphate (P) starvation. We then ask whether mortality patterns can be related to the evolutionary history of each strain, the geographic location and environmental conditions where it was isolated from, or the copy number of specific acquisition genes. We observed diverse decline patterns across starvation conditions and strains, identifying three differential features: maximum culture fluorescence, the number of fluorescence peaks during the decline stage, and the decline rate. Based on these features, each strain was categorized as being more sensitive to either nitrogen starvation or phosphorus/co-starvation. High light (HL) strains are more sensitive to N starvation, whereas other facets of the strains' evolutionary or ecological origin were not correlated with mortality features. Surprisingly, the number of genes known to be involved in either N or P acquisition in each genome was not correlated with starvation sensitivity. Rather, genes involved in DNA damage repair were associated with N sensitivity to starvation, especially in HL strains, whereas genes related to protein quality control were more abundant in LL strains and associated with P/co starvation sensitivity. These findings reveal a previously unrecognized diversity in the dynamics of starvation and mortality across closely related *Prochlorococcus* strains, potentially driven by differences in the responses to DNA and protein damage.

Keywords: Mortality, phytoplankton, starvation.

Background

Microorganisms are the most abundant life form on Earth, inhabiting environments from the guts of humans and animals to marine ecosystems. Their metabolic diversity, coupled with the capacity to rapidly reproduce under diverse environmental conditions, makes them key drivers of the biogeochemical processes essential to life. However, microbes also die. Death might be due to ecological processes, such as predation and phage infection, or from "natural" causes, including oxidative stress, elevated temperature, and nutrient depletion (Pérez et al., 2016; Shoemaker et al., 2021; Zimmerman et al., 2020). While death due to predation or phage infection has been intensively studied (e.g. (Chevallereau et al., 2022; Jousset, 2012; Pernthaler, 2005; Rønn et al., 2002; Seed, 2015)) much less is known about natural mortality, yet the rate at which bacterial cells die, and potentially how they die, can have important biogeochemical consequences. This is because the release of cellular contents from dying cells provides organic matter that can be recycled within microbial communities ("the microbial loop"; for examples from the marine environments see (Azam et al., 1983; Hansell et al., 2009; Moran et al., 2016)). In aquatic systems, cells that die but are neither internalized (eaten) by other organisms nor immediately lysed may also influence aggregation and vertical flux, contributing to the export of organic carbon and nutrients to deeper waters (e.g. (Bar-Zeev et al., 2013; Kahl et al., 2008)).

These fluxes are central to the regulation of atmospheric CO_2 and the transfer of bioavailable material to deeper ecosystems (Azam & Malfatti, 2007).

One potential cause of microbial death is starvation (Bergkessel et al., 2016; Lever et al., 2015). Different microbes respond differently to starvation, for example, in *E. coli*, slower growth under carbon starvation enhances survival (Biselli et al., 2020), while *B. subtilis* forms stress-resistant spores under nutrient limitation (Chubukov & Sauer, 2014). Different stresses also interact, leading to trade-offs; for instance, *E. coli* cells selected for UV resistance also show increased starvation tolerance but become more vulnerable to salinity changes (Goldman & Travisano, 2011). Despite these observations, natural mortality patterns in common environmental microbes, and specifically in marine bacteria, and how mortality changes with ecological and evolutionary contexts, remain poorly understood.

Here, we characterize the dynamics of starvation and mortality in *Prochlorococcus*, a globally abundant marine cyanobacterial clade, which is the numerically dominant primary producer in oligotrophic regions (Biller et al., 2015). *Prochlorococcus* haves the smallest genomes of all primary producers, providing a model for the simplest biological system capable of living off sunlight and inorganic nutrients (Biller et al., 2015; Coleman & Chisholm, 2007; Partensky & Garczarek, 2010). The *Prochlorococcus* "federation" comprises many diverse lineages, each equipped with its unique set of genes allowing it to thrive in its oceanic niche (Biller et al., 2015). These lineages are typically divided into two main categories: high-light-adapted strains (HL), that grow at the ocean surface, where nutrients are scarce, and low-light-adapted strains (LL) found in deeper waters where nutrients are more abundant. All *Prochlorococcus* strains (HL and LL) share ~1,000 core genes for essential housekeeping functions (Kettler et al., 2007), with thousands of additional "flexible" genes driving lineage-specific adaptations, giving rise to at least six ecotypes (HLI–HLVI, LLI–LLVII) (Biller et al., 2015b; Kettler et al., 2007; Partensky et al., 1993a). Differences in responses to environmental stresses, such as nutrient stress or co-stresses, can be understood, at least in part, through this core-versus-flexible gene content. Specifically, about 92 genes have been shown to be involved in the response of several *Prochlorococcus* strains to N and P limitation or starvation and these genes have been used as biomarkers to infer nutrient limitations in the ocean (Berube et al., 2015, 2019; Martiny et al., 2006; Ustick et al., 2021).
While the short-term response to starvation has been characterized extensively in *Prochlorococcus*, little is known about the subsequent, mostly unexplored stages, of long-term starvation. Previous studies have shown that carbon- and nitrogen-starved cells die rapidly, and that the rate of mortality is reduced when cells interact with heterotrophic *Alteromonas* bacteria (Coe et al., 2021; Roth-Rosenberg et al., 2020; Weissberg et al., 2023). However, to what extent the dynamics of starvation and mortality differ between different stressors, and whether they are related to the evolutionary and ecological history of different strains, remain unknown. We posit that a better understanding of starvation and mortality in *Prochlorococcus* could help understand basic aspects of cell biology and physiology in a "simple" model bacterium, and could also provide constrains on the rates of mortality under natural conditions. Indeed, the fraction of *Prochlorococcus* cells in the ocean dying from "natural" sources (i.e. neither phage predation nor grazing) is unknown, but estimates range from less than 1% to more than half (e.g. (Agustí & Llabrés, 2007; Beckett et al., 2024). To begin addressing these questions, we cultured fifteen axenic *Prochlorococcus* strains, isolated from different oceanic regions and representing broad genetic diversity, under conditions where cessation of growth and subsequent mortality are due to N and P starvation. We show that these strains exhibit diverse "death phenotypes" which can be partly quantified using specific features of the "growth and death" curves – the maximum culture fluorescence, the number of fluorescence peaks during the decline stage, and the decline rate. We propose that these features are associated primarily with the HL-LL divide and less with the specific ecological and evolutionary history of each strain. Surprisingly, these features were also not associated with known N

and P acquisition genes, but rather with genes associated with DNA and protein quality control and repair.

Material and methods

Experimental design.

Fifteen axenic strains of *Prochlorococcus* were maintained under constant light (22-25 μmole photons m^{-2} s^{-1}) at a temperature of 22 °C. Four distinct nutrient conditions were included; replete media (Pro99), low nitrogen (lowN), and two variations of low phosphate media with differing concentrations (lowP). The axenic status of the cultures was examined before the experiment by introducing 1 ml of the culture into 15 ml of ProMM (Moore et al., 2007).

The experimental design is composed of four sets, each with MED4 and MIT9312 as controls. The experiment unfolded in two stages. The first stage, spanning a week, involved the transfer of cultures from replete media to the diverse nutrient conditions, aiming to prevent carryover. Prior to initiating stage two, cell abundance was quantified using flow cytometer and the initial concentration was set to 2×10^6 *Prochlorococcus* cells per ml.

Daily monitoring of the strains' growth was conducted by measuring fluorescence (FL- ex440; em680) using a Fluorescence Spectrophotometer (Cary Eclipse, Varian) as a proxy for cell density. Cultures were sustained until all reached a fluorescence level of less than 0.1, the experimental duration extended to 90 days post-setup (stage two).

PCA ordination. PCA ordination was run on the growth curves. The fluorescence measurements were standardized via standard scalar by subtracting the mean and dividing by the standard deviation. Ordination was computed via principal component analysis (PCA). Data preprocessing was done by pandas (2.2.3). Scaling and PCA was done using sklearn (1.6.0).

Calculating specific growth and death characteristics:

Growth rates. Growth rates were calculated as described (Weissberg et al., 2023).

Normalized FL_{max}. The strains used are categorized into different clades and ecotypes, each characterized with a unique maximum yield. Therefore, for comparative purposes, the maximum FL values were normalized with respect to the strains' individual maximum FL, observed under replete conditions (Pro99).

Peak count. The peak count was determined by applying the 'find_peaks' function from the SciPy package in python (1.15.0) with specified parameters: height=0.15 and threshold=0.05. A total of 18 distinct parameter combinations were systematically tested, including an additional parameter- distance, which was set to none. The optimal parameter set, yielding minimal noise, was selected. The 'max' function, a built-in component, was integrated alongside the 'find_peaks' function to guarantee the identification of the highest peak in all instances.

D-value. The D-value is defined as the time required to reduce cell population by 90% (Pruitt & Kamau, 1993). Here, the systematic error (0.15) was subtracted from the FL_{max} to determine the D-values from the maximum growth. T_{max} is defined as the day the cultures reached their maximum yield, $T_{10\%}$ is the time taken to reach 90% decline.

$$\text{FL at } T_{10\%} = (FL_{max} - 0.15) \times 10\%$$

$$D - value = T_{max} - T_{10\%}$$

To test how growth rate, MaxFL, peak count, and D-values differ within each strain across various media, the ordinary least squares (OLS) method from the statsmodels package in Python (v0.14.4) was used. Multi-test correction was performed using t_test_pairwise() with the Benjamini–Hochberg method to adjust for multiple comparisons between media types.

Testing correlation between environmental conditions, N/P acquisition genes, COGs and starvation sensitivity.

Environmental conditions: We performed an extensive literature search to obtain the environmental conditions measured when each *Prochlorococcus* strain was isolated. When such data was unavailable, we contacted the authors. Nevertheless, data for certain strains were unavailable, therefore the World Oceanic Atlas (Reagan et al., 2024), which provides interpolated data, was used for obtaining phosphate, nitrate and temperature measurements specific to the isolation origin and timing for each strain (Table 2, Supplementary Figure S5).

Data were collected from monthly records, selecting the closest location to the actual isolation site. Measurements were taken within a 10-meter range on either side of the isolation depth (e.g., if a strain was isolated at 15m, data were collected from 5–25m). The average was then calculated for this range. No significant difference was observed between the data from the WOA and those available during the strain isolation (Supplementary Figure S5).

N and P acquisition genes: To test whether starvation sensitivity was related to the genetic capacity to take up N or P resources, we selected 50 N-related genes associated with the utilization of ammonium, nitrate, nitrite, urea, amino acid and molybdopterin biosynthesis (Berube et al., 2015; Díez et al., 2023a; Martiny et al., 2009). Additionally, 47 P-acquisition genes were chosen for their role in phosphate, phosphite, phosphonate and phosphoanhydride metabolism (Doré et al., 2023; Kathuria & Martiny, 2011; Martínez et al., 2012; Martiny et al., 2006; Saunders & Rocap, 2016; Scanlan & West, 2006). The presence and frequency (measured by copy number) of these genes was assessed in each strain using the phyletic pattern data in the Cyanorak (Garczarek et al., 2021) (Dataset 2).

COGs across genomes: To obtain COGs data, the CyanoRak database was accessed, which provides GFF files describing the genomes of the selected *Prochlorococcus* strains, apart from MIT1314 and MIT1327. COG numbers were extracted from these files using a Python script (3.12.8).

MIT1327 was not fully annotated; however, COGs were retrieved from its GF file provided by the CyanoRak team. MIT1314 was excluded from this analysis as it has not yet been annotated.

To evaluate significant differences between measurements (e.g., environmental measurements, acquisition genes or COG copy numbers) between two groups (such as HL/LL or N/other sensitive), Mann-Whitney U test was applied to compare the distributions of the data between the groups for each measurement. To account for multiple comparisons, multiple testing correction was performed using the Benjamini-Hochberg procedure.

Results and discussion

Diverse growth and death patterns across strains and starvation types.

Prochlorococcus is usually cultured in media where the main macronutrients N and P are balanced, i.e. provided at the ratio found in biomass (the Redfield Ratio, (Redfield, 1958)). Under such conditions the limiting factor leading to cessation of growth is unknown and can include not only nutrient starvation but also carbon limitation and/or toxicity due to metabolic byproducts (Christie-Oleza et al., 2017; Moore et al., 2007). Therefore, to measure how N or P starvation affect *Prochlorococcus* growth and death, we modified the commonly used "Redfield Ratio" Pro99 medium by reducing NH_4^+ concentrations 8-fold ("lowN", (Grossowicz et al., 2017)) or PO_4^{3-} concentrations (8- or 50-fold "lowP", (Krumhardt et al., 2013)). We chose two different concentrations of PO_4^{3-} because *Prochlorococcus* are known to significantly reduce their P requirements (e.g. through the production of sulfolipids (Van Mooy et al., 2006)), leading to highly variable N:P ratios (Moreno & Martiny, 2018), and we hypothesized that an 8-fold reduction might not induce P starvation in all strains. These media were used to culture 15 axenic *Prochlorococcus* strains, which were isolated from different oceanic locations and represent much of the genetic diversity of this globally abundant clade (Figure 1A, B and Table 1).

The growth and death of each strain were followed over 90 days by measuring bulk culture fluorescence, a proxy for cell growth, resulting in a dataset of 252 growth curves. The "growth-and-death" curves were highly reproducible both between biological triplicates (Figure 1C, Supplementary Figure S1) and between independent experiments with two strains (MED4 and MIT9312, Supplementary Figure S2 and Supplementary Text S1). Clear differences in the shape of the growth-and-death curves under N and P starvation, as well as between nutrient starved conditions and the "Redfield Ratio" Pro99 medium, were observed both within and between strains (Figure 1C, Supplementary Figure S1 and S3). Most strains showed no significant differences in exponential growth rates under lowN (1:8) and lowP (1:8) conditions, with similar growth observed in 80% of cases. However, under more severe P starvation (1:50), most strains exhibited a reduced growth rate. Nevertheless, the main difference was visually seen between the different nutrient starvation types, and between the different strains, was in the mortality stage, consistent with previous studies (Weissberg et al., 2023).

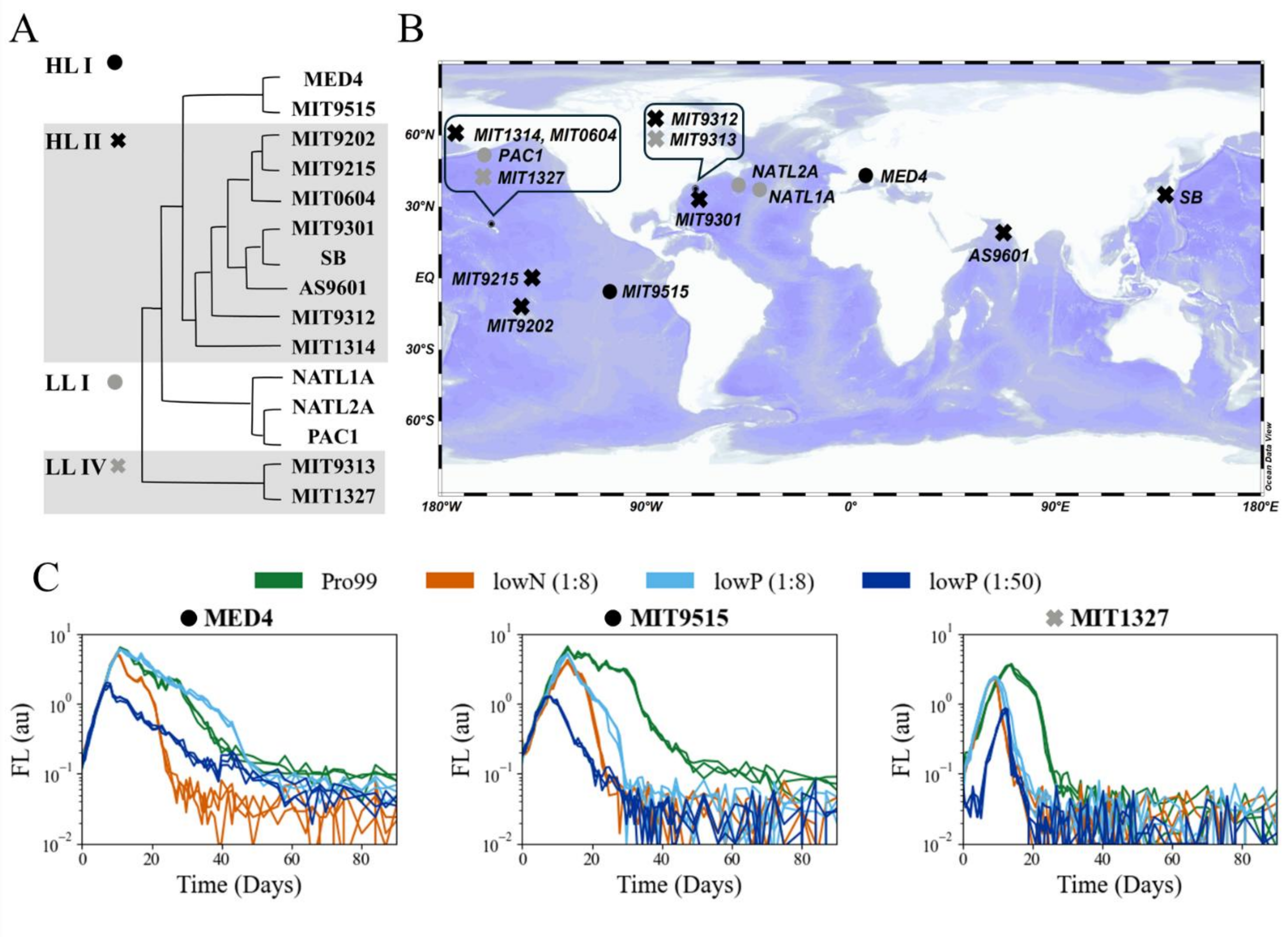

Figure 1: the characteristics of the 15 strains used, and examples of growth and-death curves under multiple nutrient conditions.

A. Schematic phylogenomic tree (based on (Becker et al., 2024; Biller et al., 2014a)) illustrating the evolutionary diversity of the 15 strains used; black and grey represent HL and LL lineages, respectively, while shapes (circle and "X") indicate distinct ecotypes. B. A world map highlighting the isolation location of each strain. Note that multiple strains may have been isolated from the same location. C. Growth-and-death curves for three selected strains (see Supplementary Figure S1 for all curves); the x-axis represents time in days, and the (logarithmic) y-axis represents bulk culture chlorophyll autofluorescence (FL, arbitrary units). Individual lines represent biological triplicates. Colors represent the media under which cultures grew, while shapes above each plot denote clades as shown in panel A. Highly variable fluorescence under 10^{-1} au is considered instrument noise.

Table 1: the axenic *Prochlorococcus* strains used in this study, with information on when and where they were isolated, environmental conditions at the time of isolation (from the World Oceanic Atlas), and sum of known N and P acquisition genes.

Temperature (C)	Nitrate (μmol/kg)	Phosphate (μmol/kg)	P acquisition genes	N acquisition genes	Strain reference
13.34	2.36	0.11	34	22	(Moore et al., 1995)
26.81	2.84	0.4	14	11	(Rocap et al., 2002)
26.77	0.78	0.05	17	18	(Moore & Chisholm, 1999)
25.61	5.73	0.76	16	19	(Moore & Chisholm, 1999)
20.04	2.85	0.18	20	39	(Biller et al., 2014b)
20.05	0.36	0.03	37	19	(Rocap et al., 2002)
23.37	2.15	0.17	24	32	(Shimada et al., 1995)
26.52	2.9	0.63	16	19	(Shalapyonok et al., 1998)
19.54	1.47	0.36	27	22	(Moore et al., 1998)
20.34	0.93	0.13	20	18	(Becker et al., 2019)
16.89	0.87	0.13	28	25	(Partensky et al., 1993b)
17.34	3.17	0.11	28	25	(Scanlan et al., 1996)
21.85	0.08	0.08	19	31	(Parpais et al., 1996)
20.34	0.93	0.18	22	25	(Moore et al., 1998)
19.54	1.47	0.36	17	23	(Cubillos-Ruiz et al., 2017)

Strain	Clade	Isolation date	Isolation location	Coordinates	Depth (m)	Genome size	GC%
MED4	HLI	January 1989	Mediterranean Sea	43°N 6°E	5	1,657,991	30.8
MIT9515	HLI	June 1995	Tropical Pacific	5.7°S 107.1°W	15	1,704,176	30.8
MIT9202	HLII	September 1992	Tropical Pacific	12°S 145.4°W	79	1,691,453	31.1
MIT9215	HLII	October 1992	Equatorial Pacific	0° 140°W	5	1,738,790	31.1
MIT0604	HLII	May 2006	North Pacific	22.75°N 158°W	175	1,780,061	31.2
MIT9301	HLII	July 1993	Sargasso Sea	34.8°N 66.2°W	90	1,641,879	31.3
SB	HLII	October 1992	Western Pacific	35°N 138.3°E	40	1,669,973	31.5
AS9601	HLII	November 1995	Arabian Sea	19°N 67.10°E	50	1,669,886	31.3
MIT9312	HLII	July 1993	Gulf Stream	37.5°N 68.2°W	135	1,709,204	31.2
MIT1314	HLII	June 2013	North Pacific	22.75°N 158°W	150	1,704,447	31.2
NATL1A	LLI	April 1990	North Atlantic	37.4°N 40°W	30	1,864,731	35.0
NATL2A	LLI	April 1990	North Atlantic	39°N 49.3°W	10	1,842,899	35.1
PAC1	LLI	April 1992	North Pacific	22.75°N 158°W	100	1,842,113	35.1
MIT9313	LLIV	July 1993	Gulf Stream	37.5°N 68.2°W	135	2,410,874	50.7
MIT1327	LLIV	June 2013	North Pacific	22.75°N 158°W	150	2,587,389	50.3

Quantitative features of culture decline.

While the growth phase of most cultures could be described by a single, exponential growth rate, the stationery and decline phases are much more complex, with multiple peaks, troughs and decline stages (Figure 1C, Supplementary Figure S1). We therefore defined three quantitative features, each describing different aspects of the transition to nutrient starvation and subsequent mortality: the maximum fluorescence, number of fluorescence peaks, and the time it took the cultures to decline to

10% of maximum fluorescence (decimal reduction time, D-value, (Pruitt & Kamau, 1993)). Each feature captures a distinct aspect of cellular death that is differently affected by starvation conditions (for statistical results see Supplementary Table S2 and for measurements see Dataset 1).

Maximum fluorescence yield- Max_{FL}. Bulk culture chlorophyll Fluorescence (FL) is an easily measurable, non-invasive parameter that, generally, is correlated with cell abundance during growth and decline stages (Weissberg et al., 2023, 2025). However, interpreting fluorescence maxima as direct measures of cell density requires caution, as chlorotic cells emerge under starvation. These cells are characterized by low chlorophyll autofluorescence and can exhibit reduced FL without necessarily indicating a decrease in cell abundance (Roth-Rosenberg et al., 2020). Our findings indicate that Max_{FL} is strongly influenced by growth conditions, with the replete medium consistently producing the highest yield across nearly all strains. Interestingly, in 40% of cases, growth in lowP (1:8) resulted in a Max_{FL} comparable to Pro99. This suggests that an 8-fold reduction in P may not be sufficient to induce P starvation in these strains, with cessation of growth potentially due to other factors such as the accumulation of waste products. Additionally, while lowP (1:50) reduced Max_{FL} compared to lowP (1:8), the decrease was not as large as the ~six-fold reduction expected from a simple linear relationship between available PO_4^{3-} and biomass. In contrast, Max_{FL} was lower in all strains grown in lowN compared to Pro99, although the reduction was also generally less than the expected eight-fold decrease compared to Pro99 (Max_{FL} Figure 2B, see also (Grossowicz et al., 2017)). Taken together, these results suggest that lowN (1:8) and lowP (1:50) induced N and P starvation, respectively, but that quantitatively assessing the impact of starvation on yield based on bulk culture autofluorescence may be complicated by varying physiological responses (e.g., chlorosis and, in the case of Pro99, additional mortality factors such as the exudation of toxic byproducts, (Grossowicz et al., 2017)).

Peak count. As the population declines, multiple peaks and troughs are observed in the growth-and-death curves, and these patterns are reproducible across replicates and conditions (Figure 1, Figure 2B, Supplementary Figure S1). These peaks may reflect changes in physiological properties such as fluorescence per cell, or phases where cell abundance actually increases. In *E. coli*, peaks and troughs in the growth and death curves can be due to the cells beginning to catabolize amino acids, carbohydrates, lipids, and even DNA from dying cells (Finkel, 2006; Finkel & Kolter, 2001). This process continues until all viable cells are exhausted, and (primarily at later stages, after most cells have died) may be accompanied by the emergence of genetically distinct lineages with mutations conferring growth advantage under stationary phase (the "GASP" phenotype, (Finkel, 2006). In *Prochlorococcus*, we calculated the number of peaks from the point of Max_{FL} and onwards, observing that most of the peaks occur in the stationary or early decline phases, and mostly in the Pro99 media (Figure 2, Supplementary Figure S1). We do not currently know whether they represent changes in per-cell fluorescence or in cell numbers but given that these peaks occurred relatively early and when FL was high, it is less likely that they represent a true GASP phenotype. In contrast to most strains, three strains (PAC1, MIT9313 and MIT1327) did not exhibit multiple peaks under nutrient-deplete condition, and in MIT1327 they were not observed even in Pro99 (Figure 2B). What drives these changes in culture fluorescence, and what molecular processes define them; requires further investigation.

D-value. The D-value, a widely used metric in microbiology, refers to the time required for 90% of the cells in a culture to die (Pruitt & Kamau, 1993). A large D-value therefore reflects a slower decline rate and prolonged survival. Across almost all strains and media, the Pro99 replete medium exhibited the highest D-value, whereas N-deplete conditions led to the fastest decline (smallest D-value) in 8 of the 15 strains. This contrasts with the observation that lowP (1:50) led to the lowest yield in all strains, suggesting that the processes involved in cessation of growth and in death may not be the same. It is tempting to speculate that this represents reallocation of resources from growth to survival when cells sense P-starvation. Interestingly, a positive correlation (r = 0.71) was observed between the peak count and D-value, suggesting that dynamic changes during stationary stage (i.e. changes in per-cell fluorescence or in cell numbers, whether due to nutrient starvation or another stressor) may play a significant role also in determining death rate (Supplementary Figure S4).

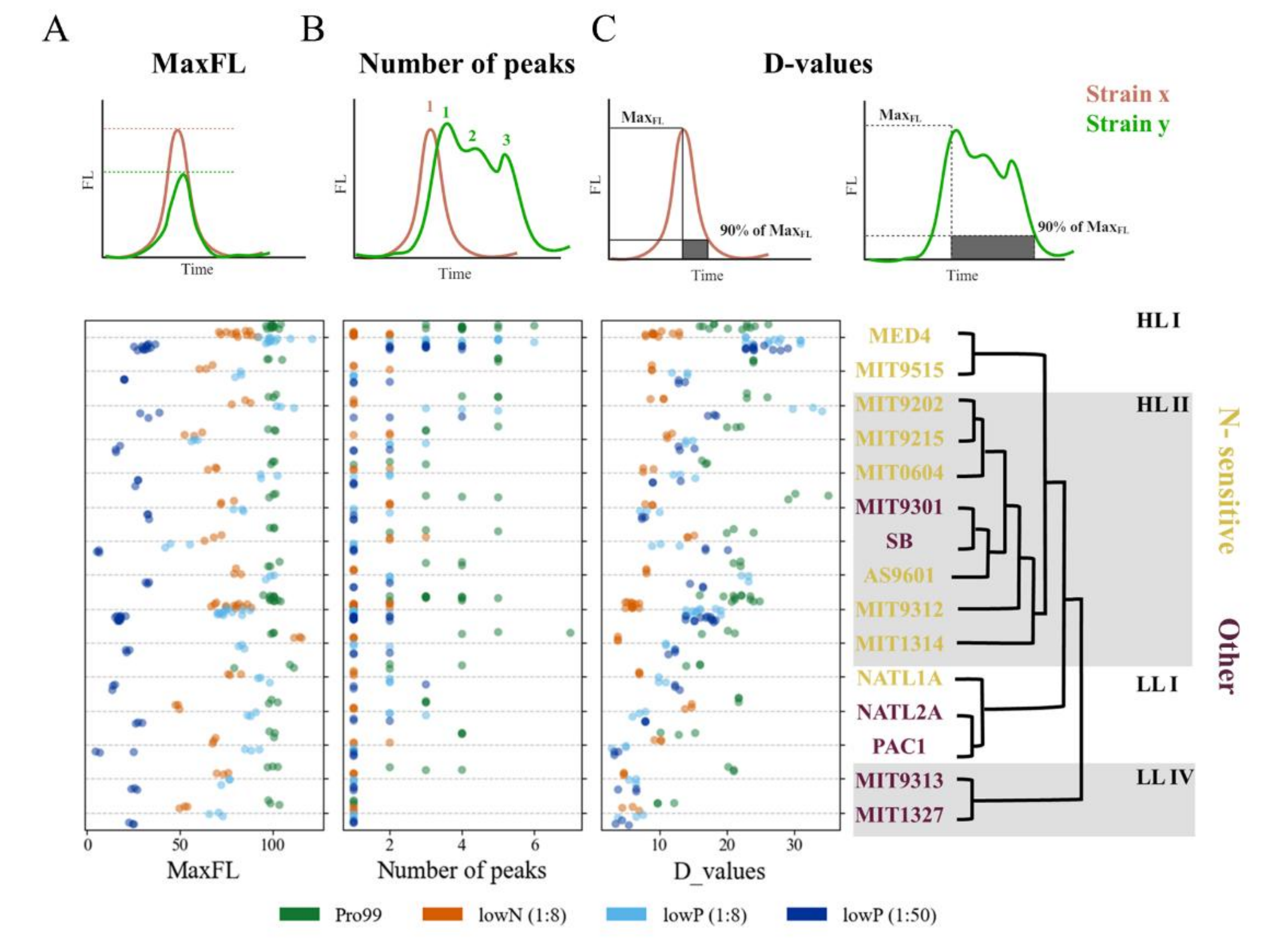

Figure 2: Features describing the culture decline stage.

A. Max_{FL}; B. Peak count; C. D-values. Circles represent biological triplicates. MED4 and MIT9312 served as controls for multiple experiments, resulting in a higher number of data points (for more details, see Methods). Jitter was introduced to improve the visualization of overlapping points. Strain names are colored according to their sensitivity type, which is determined by the D-value (see text). A lower D-value indicates a faster decline in a given condition, reflecting greater sensitivity.

Sensitivity to starvation is associated with the HL-LL division in Prochlorococcus

We next asked whether the different patterns of mortality can be related to the evolutionary or ecological background of the different *Prochlorococcus* strains. The phylogeny of the strains (figure 1A), and particularly the differentiation between HL and LL adapted clades, is associated with evolutionary differences in many traits such as genome size, metabolic complexity and, potentially, the capacity to utilize diverse organic compounds (Table 2, (Biller et al., 2015)). An alternative hypothesis is that the patterns of mortality represent adaptation of each *Prochlorococcus* strain to the specific local environmental conditions present when and where it was isolated (Ustick et al., 2021). For instance, the North Atlantic has lower phosphate concentration in surface waters compared to the North Pacific, which is mirrored by the number of genes encoded in the genomes in strains from the region that are associated with N or P uptake (Ustick et al., 2021). To test these two hypotheses, we first divided the strains into those that are more sensitive to N starvation and those which are either more sensitive to P starvation or equally sensitive to N and P (which we termed "other" sensitive). We based this classification on the D-value, as it provides a more direct and interpretable measure of cell death compared to Max_{FL} and peak count (see Supplementary Table S3, also for classification based on peak count and Max_{FL}).

LL adapted strains have larger genomes and a higher predicted capability to utilize various organic resources such as amino acids and nucleotides and thus grow mixotrophically (using organic resources rather than only inorganic ones, (Muñoz-Marín et al., 2020; Wu et al., 2022; Yelton et al., 2016)). We therefore expected these strains to be better at utilizing organic matter released from dying cells ("necromass", (Shoemaker et al., 2021)), resulting in slower decline (higher D value), and potentially more peaks of "re-growth". However, while the most apparent difference in sensitivity was indeed between the HL and LL adapted clades, it was the HL adapted strains that declined more slowly under P starvation (higher D-values, Figure 3A), suggesting they are more adapted to this stress (Figure 3A). In contrast, no difference was observed in the rate of decline under N starvation.

Apart from the HL/LL division, the ecological and evolutionary history of the *Prochlorococcus* strains had no clear relationship with the sensitivity to N or P starvation (Figure 3B). There was no correlation between starvation sensitivity and the environmental conditions (temperature, nitrate or phosphate concentrations) when and where each strain was isolated, estimated using the World Oceanic Atlas (WOA) ((Reagan et al., 2024), also see Supplementary Text S2). Moreover, no such correlation was observed when the average seasonal or annual environmental conditions were considered (Supplementary Figure S5, Supplementary Table S5 and Supplementary Text S2). Finally, starvation sensitivity did not correlate with the number of genes associated with N or P uptake ((Díez et al., 2023b; Doré et al., 2023; Martiny et al., 2006, 2009), Figure 3B, see materials and methods for details). The only correlation observed was with the G/C content of the genomes, which reflects the HL/LL division (Moore et al., 1995). Taken together, these results suggest that while the deep evolutionary history of the *Prochlorococcus* clade is somehow related to the sensitivity to different types of nutrient starvation, the short-term ecological history of each strain, or the (known) genes for N and P acquisition encoded in its genome, are unlikely to drive the distinct death patters observed (Figure 3A, Supplementary Table S4).

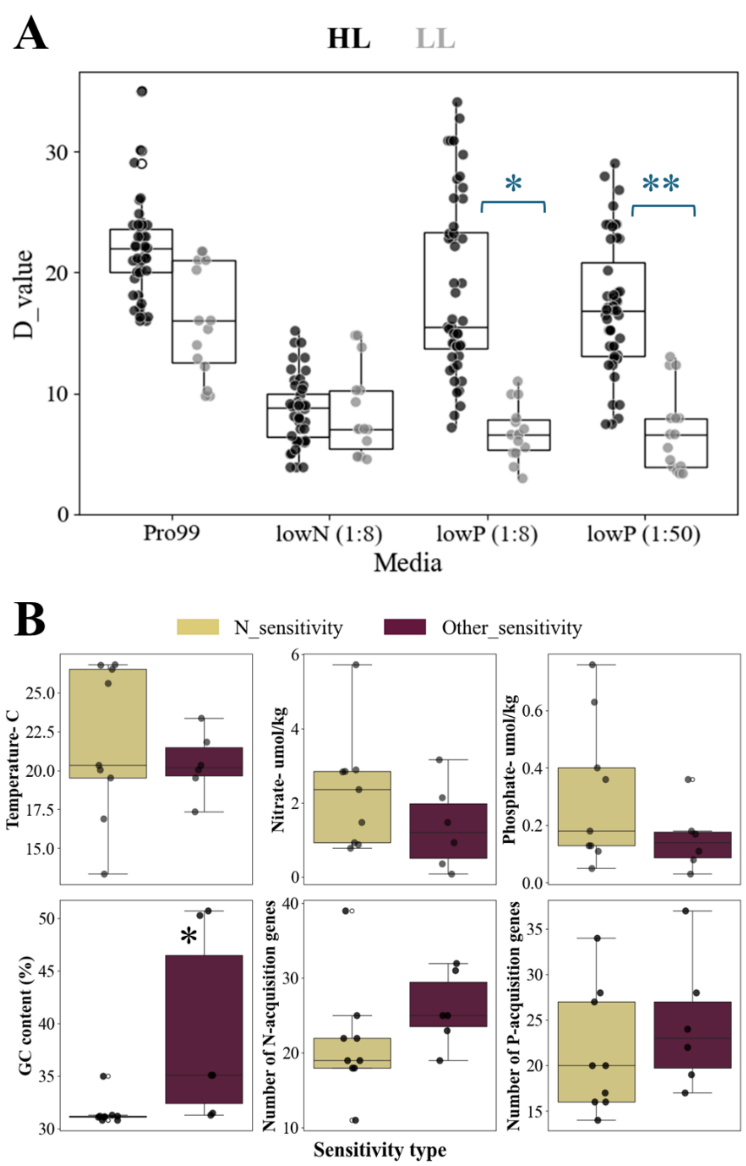

Figure 3: Evolutionary history partially correlates with culture decline whereas ecological history (environmental conditions when and where strains were isolated) does not.
A. D-values are higher under P starvation in HL strains, asterisks indicate statistically significant differences. B. Environmental conditions when and where strains were isolated, G/C content and number of N and P genes known to be associated with nutrient acquisition in N- and other-sensitive strains, as classified using the D-value. For results based on Max FL and peak number see Supplementary Table S4.

Identifying genes that are potentially associated with starvation sensitivity.

We next searched for genes that are differentially distributed between N and "other"-sensitive strains, and that could hint at the molecular mechanisms underlying the differential mortality patterns. To do so, we examined the abundance of genes belonging to different Clusters of Orthologous Genes (COGs) across the *Prochlorococcus* genomes studied, using CyanoRak (Garczarek et al., 2021). There are 1,380 COGs in the *Prochlorococcus* pan genome, of which 670 varied in gene number between strains. 112 of these differed in number between HL/LL strains or N- and "other" sensitive ones (Figure 4A).

Although some COGs appeared to be associated with N or "other" sensitivity, the differences in gene count within each COG were relatively small. In fact, no significant associations remained after applying the Mann-Whitney U test followed by the Benjamini-Hochberg correction for multiple testing. Additionally, interpretation of the results was complicated by the cross-correlation between P sensitivity and LL strains, as shown in Figure 3A. Despite this, certain patterns emerged among the COGs linked to specific types of starvation sensitivity. These patterns may offer insight into potential mechanisms underlying the observed mortality responses. Therefore, we highlight the COGs that showed significance at $p < 0.05$ without correction for multiple testing (see Table 2 and Dataset 3).

Overall, 26 COGs were associated with both starvation type and evolutionary history (the HL/LL division), and another 15 were only associated with starvation sensitivity (Figure 4A, Table 2). The pattern that was most clearly associated with sensitivity to N starvation was an increased number of genes involved in DNA damage control, namely DNA gyrase (COG0188), helicase/nuclease (recB, COG1787), endonucleases (COG3727) and DNA ligase (COG1793), (Table 3, Figure 4B). One possible interpretation of these results is that an increase in the copy number of these genes sensitizes the strains to N starvation, but we propose an alternative thought, which is that this increase represents a mechanism to better deal with P starvation (i.e. these genes represent lower sensitivity to P starvation in HL strains, rather than higher sensitivity to N starvation). In support of this interpretation, the enrichment of DNA damage repair genes in HL strains aligns with their ecological niche, as these strains predominantly inhabit the ocean's surface, where nutrient availability is limited and exposure to UV radiation and reactive oxygen species (ROS) is high. These environmental stressors are strongly linked to DNA damage (Cabiscol Elisa et al., 2000; Sinha & Häder, 2002), necessitating an enhanced capacity for DNA repair. Further support is provided by the observation that two HL strains, SB and MIT9301, which were "other" sensitive rather than N-sensitive, had lower numbers of the COGs associated with DNA damage compared with other HL strains.

In contrast, strains more sensitive to P/co starvation (and thus less sensitive to N starvation) exhibited a greater abundance of genes associated with protein quality control including chaperones (COG2214, COG0443, and COG0484) and a protease (COG0330), (Table 3, Figure 4B). These genes were also more abundant in LL strains. We speculate that the association of these genes with lower sensitivity to N starvation is because N starvation leads to amino acid depletion, causing nascent polypeptides to become "stuck" on the ribosomes (Betney et al., 2010). Dealing with starvation requires chaperones and proteases which can prevent protein aggregation, assist in refolding misfolded intermediates, or direct unsalvageable proteins for degradation (Mayer, 2021). Thus, a higher number of such genes could result in less protein misfolding and aggregation, reduced cellular toxicity, and ultimately slower death under N starvation (and a higher relative sensitivity to P/co starvation). However, why these functions are also more abundant in LL strains remains unclear.

Genes related to mixotrophy and/or alternative nitrogen sources, such as amino acid metabolism (COG1454) and a urease subunit (COG2370), were more abundant in P/co- sensitive strains, but also in LL strains. As noted above, while in principle a higher number of such genes could suggest better resistance to N starvation through mixotrophy (Yelton et al., 2016), LL strains and HL strains die at the same rate under N starvation (Figure 3A), and thus increased mixotrophy does not seem to be a strong adaptive strategy to cope with this stress. Genes related to lipopolysaccharides, exopolysaccharides, biofilm attachment, energy production, translation, transcription, and signal transduction were also differentially associated with either N or P/co sensitivity, but these associations were less consistent (Table 3), and we currently cannot propose an explanation for them.

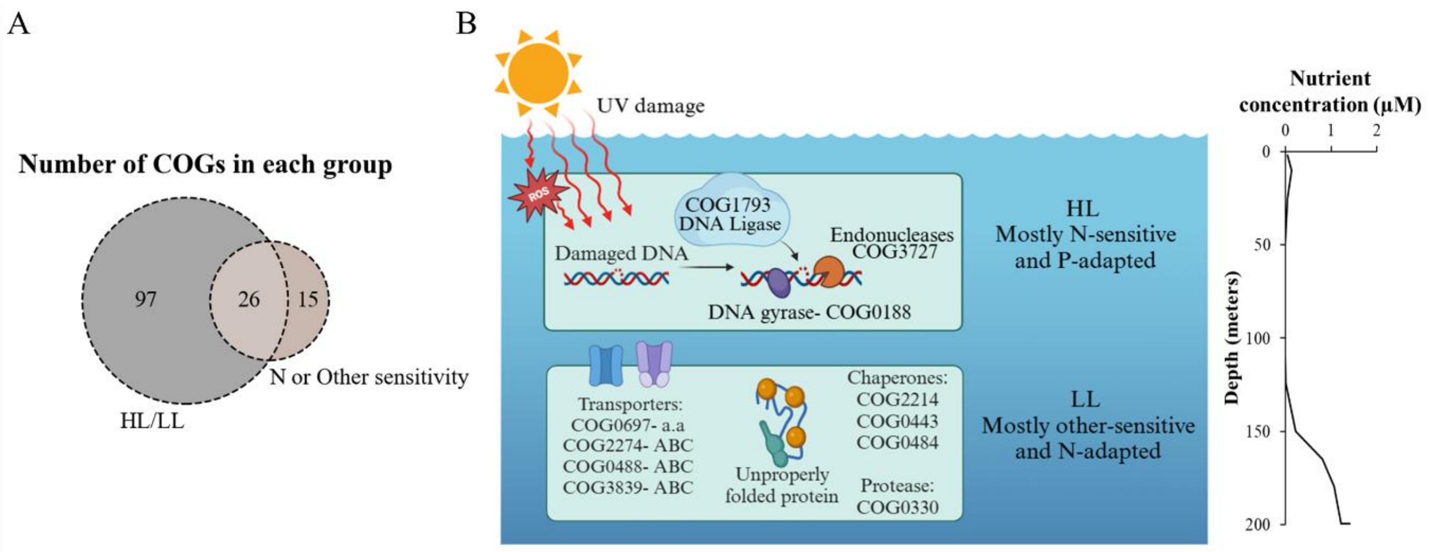

Figure 4: Proposed cellular mechanisms underlying starvation sensitivity in *Prochlorococcus* and their division between HL and LL clades.
A. The number of COGs differing between HL and LL strains and/or between N and co-sensitive ones ($p<0.05$, uncorrected for multiple hypothesis testing). B. An illustration showing key cellular mechanisms underlying death in *Prochlorococcus* on the left, and a nutrient profile from the Eastern Mediterranean on the right, illustrating nutrient concentrations in regions inhabited by HL and LL strains.

Table 2: COGs potentially differ in abundance between N and other sensitive strains.

Proposed process	**Annotation**	**Protein**	**COG**	**p-value (uncorrected)**	**More abundant in strains sensitive to**	**More abundant in strain belonging to**
Quality control-DNA level	Gyrase and DNA topoisomerase	GyrA	COG0188	0.012	N	HL
	Mrr, RecB and HJR activity	Mrr, RecB and HJR	COG1787	0.047	N	-
	very short path repair (Vsr) endocluases	Vsr	COG3727	0.024	N	LL
	DNA ligase I	CDC9	COG1793	0.047	N	LL

	ArsR family	smtB/ArsR	COG0640	0.024	Other	HL
	type IV methylase	McrA	COG1403	0.014	N	HL
Quality control-Protein level	Hsp member- CbpA protein- with DnaJ like domain	CbpA	COG2214	0.041	Other	LL
	Hsp member- Hsp70	DnaK	COG0443	0.034	Other	LL
	Hsp member- DnaJ chaperone	DnaJ	COG0484	0.047	Other	LL
	Protease- HflB (FtsH) modulator	HflC	COG0330	0.041	Other	LL
	a pseudouridylate synthesase- site specific	RsuA	COG1187	0.041	Other	LL
Amino acid and mixotrophy related biosynthesis	Amino acids as nitrogen sources-threonine dehydrogenase	EutG	COG1454	0.013	Other	LL
	hydrogenase/ urease accessory protein HupE/UreJ protein	HupE	COG2370	0.024	Other	-
	Asparagine synthetase B	AsnB	COG0367	0.028	N	-
	Chorismate synthase	AroC	COG0082	0.041	Other	LL
	ABC-type bacteriocin/lantibiotic exporters	SunT	COG2274	0.042	Other	LL
	type II secretory pathway	PulO	COG1989	0.045	Other	LL
	membrane-fusion protein	AcrA	COG0845	0.042	Other	LL
	ATPase components of ABC transporters	Uup	COG0488	0.047	Other	LL
	Carbohydrate transport and metabolism, Amino acid transport and metabolism	RhaT	COG0697	0.028	Other	LL
	ABC transporter involved in maltose transport	MalK	COG3839	0.041	Other	LL
LPS	CMP-KDO synthetase	kdsB	COG1212	0.037	Other	-
	Carbohydrate transport and metabolism, Amino acid transport and metabolism	prop	COG0477	0.009	Other	LL

	carbohydrate transport and metabolism	BglX	COG1472	0.041	Other	LL
	Lipopolysaccharide biosynthesis protein	RgpF	COG3754	0.038	Other	LL
	sugar aminotransferase	WecE	COG0399	0.048	N	HL
	poorly characterized, dehydrogenase- sugar aminotransferase fusion protein	MviM	COG0673	0.024	N	-
Membrane linker	links inner and outer membranes	TonB	COG0810	0.041	Other	LL
Exopolysaccharide	GumC/Wzc1, integral membrane protein	GumC	COG3206	0.009	N	-
	outer membrane channel, ESP are translocated by it	Wza	COG1596	0.004	N	HL
Biofilm/Attachment	CsgG protein	CsgG	COG1462	0.037	Other	-
	Large exoprotein involved in heme utilization or adhesion	FhaB	COG3210	0.029	Other	-
Energy production	Cytochrome c mono- and diheme variants	CccA	COG2010	0.048	Other	-
Translation, transcription and signal transduction	Adenylate cyclase	AcyC	COG2114	0.038	Other	-
	Acetyltransferase	PhnO	COG0454	0.039	Other	-
RTX toxins	RTX toxin-related domain		COG2931	0.012	Other	LL
Others	Dienelactone hydrolase	DLH	COG0412	0.013	Other	-
	Extracytoplasmic sensor domain CHASE2	CHASE2	COG4252	0.038	Other	-
	carbamoyl transferase		COG2192	0.011	N	HL
	lignin degradation	LigW	COG2159	0.002	N	-
	PPE-repeat proteins	PPE	COG5651	0.037	Other	-

Conclusion, caveats and prospects

Death due to nutrient starvation remains poorly understood in microbes in general, and in marine bacteria (including cyanobacteria) in particular. In this study, we investigated phenotypic features of starvation and mortality-maximum fluorescence, number of peaks, and death rates- across diverse *Prochlorococcus* strains under N and P starvation. These measurements revealed complex and strain-specific patterns, raising new hypotheses regarding the molecular mechanisms underlying the response to starvation.

The patterns of mortality due to starvation were influenced by the strains' evolutionary history (HL/LL divide), but not, as far as we could resolve with the culture collection in our hand, by their shorter-term ecological history (based on the conditions where they were isolated from or the number of previously-characterized genes involved in N and P acquisition encoded in their genomes). This is surprising, given that N and P acquisition genes have been considered as markers for microbial survival and adaptivity in nutrient-limited environments (Berube et al., 2015; Martiny et al., 2006, 2009). Furthermore, N acquisition genes have been linked to phylogeny, whereas P acquisition genes exhibit a correlation with environmental location (Berube et al., 2019; Coleman & Chisholm, 2010). We note that these mapping the evolutionary and ecological history of the strains was non-trivial, as the record of when and where many strains were isolated, and what the environmental conditions were at the time, was partial or missing. Systematically collecting and reporting such data is necessary for future work to integrate phenotypic, genomic, and environmental data.

Moreover, encoding more genes related to mixotrophy in the genome also does not influence the rate of mortality, despite our expectation that the ability to utilize more organic molecules as resources would enable LL strains to deal better with N and P starvation (Table 3, (Muñoz-Marín et al., 2020; Yelton et al., 2016)). Instead, we identified genes related to protein and DNA quality control that might be associated with the rate of *Prochlorococcus* mortality in response to N or P starvation. These associations suggest that protein homeostasis may play a key role in LL strains, while DNA repair mechanisms are particularly important for managing P-starvation, primarily in HL strains. Currently, there are no methods for gene manipulation in *Prochlorococcus* (Laurenceau et al., 2020), and therefore testing these hypotheses will either have to wait for such methods to be developed, or they will need to be tested in related cyanobacteria such as *Synechococcus*.

One key aspect of bacterial physiology not assessed in our study is the immediate physiological and biochemical response to starvation. This could include the stringent response, a mechanism whereby bacteria undergo cellular modifications that slow down transcription, translation, and DNA replication, enabling them to better survive starvation (Irving et al., 2021). Two of the identified COGs (COG0188, a DNA gyrase, and COG1787, the helicase RecB) are associated with the stringent response in *Escherichia coli*. and *Staphylococcus aureus*, through their roles in inhibiting replication by binding single-stranded DNA (Lanyon-Hogg, 2021). It is tempting to speculate that inter-strain differences in the stringent response in *Prochlorococcus* might underscore some of the differences in starvation and mortality. Future work should aim to clarify how physiology, biochemistry and gene expression change as cells transition from survival to death, which could help unravel the molecular programs that govern starvation and death in oligotrophic marine microbes.

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